

Independent exact certification of the dimension-eleven kissing records

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Abstract

The kissing number $K(d)$ is the largest number of non-overlapping unit spheres that can touch a central unit sphere in $\mathbb{R}^d$. In dimension eleven the record stood near 582 for decades and then moved three times in eighteen months, each time by an AI system, and each configuration was validated by the verifier of the group that produced it. We re-decide every publicly available witness in exact arithmetic over $\mathbb{Z}[\sqrt{2}]$ on arbitrary-precision integers, sharing no code with any producer, and confirm $K(11) \geq 604$. The three 604-point configurations have distinct exact contact counts (19,704, 22,904, 22,840), which certifies that they are pairwise non-congruent. One claim resists certification for a reason worth recording: the configuration behind the headline 604 of one platform is not published, so its row measures opacity rather than geometry. No new bound is claimed and no upper bound is touched.

1 The decision procedure

A finite set of non-zero vectors $x_1, \dots, x_n \in \mathbb{R}^d$ yields a kissing configuration exactly when the pairwise angles are at least 60° : place a unit sphere at $2x_i/|x_i|$ and all of them touch the central sphere without overlapping. The condition is invariant under scaling each vector separately, so any exact representative of each direction decides the claim, and for $i \neq j$ it reads

$$\langle x_i, x_j \rangle \leq 0 \quad \text{or} \quad 4\langle x_i, x_j \rangle^2 \leq \langle x_i, x_i \rangle \langle x_j, x_j \rangle.$$

Both alternatives are polynomial in the coordinates, so a configuration given in any real quadratic field can be decided without error. The 604-point records live in $\mathbb{Q}(\sqrt{2})$, so the certifier works in $\mathbb{Z}[\sqrt{2}]$: a number is a pair (a, b) standing for $a + b\sqrt{2}$ with a, b arbitrary-precision integers, and the sign of $a + b\sqrt{2}$ is decided by comparing a^2 with $2b^2$ when the coefficients disagree in sign. The tie $a^2 = 2b^2$ with $(a, b) \neq (0, 0)$ is impossible over the integers because $\sqrt{2}$ is irrational; the implementation raises on it, and that raise is exercised as a control. No floating-point number participates in any decision.

A decimal input such as -0.1449 is read as the exact rational it denotes, $-1449/10000$, and never as the nearest binary double. This matters in Section 3.

*Independent researcher. Contact: carlos@carlostoledo.co. This work was produced with substantial AI assistance in a certified-verification setting: constructions and exposition were proposed with the help of large language models, and every quantitative claim is decided by outward-rounded interval arithmetic or exact rational arithmetic, with adversarial controls that must fire for the certificate to be accepted. The certificates, not the models, are the evidence, and they are public and re-runnable. Machine-derived and not peer-reviewed.

2 Calibration

The instrument is calibrated at every run against two configurations whose kissing numbers are known exactly, generated from their own definitions rather than stored: the D_4 root directions give $K(4) \geq 24$, and the E_8 roots give $K(8) = 240$. The E_8 run reports 6,720 pairs at exactly 60° , which is the textbook count $240 \cdot 56/2$ for that root system and is reproduced as an equality in exact arithmetic rather than as a float comparison.

3 The dimension-eleven ladder

Theorem 1. $K(11) \geq 604$, *certified in exact arithmetic over $\mathbb{Z}[\sqrt{2}]$ from published coordinates, independently of any producer’s verifier.*

bound	witness	verdict	exact contacts
582	classical norm-four shell	CERTIFIED	22,560
593	AlphaEvolve, integer coordinates	CERTIFIED	0
594	platform rung winner, decimal coordinates	CERTIFIED	17,088
604	configuration 1, $(a + b\sqrt{2})/6$	CERTIFIED	19,704
604	configuration 2, $(a + b\sqrt{2})/6$	CERTIFIED	22,904
604	configuration 3, $(a + b\sqrt{2})/6$	CERTIFIED	22,840
604	headline claim, coordinates unpublished	NEEDS DATA	—
604 in $\mathbb{R}^{12}$	integral D_{12} lift of configuration 3	CERTIFIED	22,840

Each certified row is decided at shell norm exactly four where the producer states one, and every enclosure is an exact comparison rather than a tolerance.

Corollary 1. *The three 604-point configurations are pairwise non-congruent.*

Proof. Congruences preserve the multiset of pairwise inner products and therefore the number of pairs at exactly 60° . The three configurations have 19,704, 22,904 and 22,840 such pairs, computed as exact equalities in $\mathbb{Z}[\sqrt{2}]$, so no two can be congruent. $\square$

Remark. The rung winner at 594 is instructive about verification rather than about geometry. That platform scores submissions in fixed-precision decimal and switches to exact integer arithmetic only for integer-valued submissions, and the winning vectors are not integral. Read as the exact rationals their decimal literals denote, they are nonetheless a genuine exact witness, with 17,088 pairs at exactly 60° and every other pair strictly clear. High-precision decimal is not proof; here it happened to be reporting a true statement.

4 What could not be certified, and why that is the sharpest row

One row of the ledger is not a geometric verdict. The configuration behind the headline 604 of one platform, which the field’s reference tables credit, is not published: the public interface serves the solved 594 rung and an open 605 rung, and discussion of the 604 describes it without carrying its coordinates. The claim is very likely true. It is also, today, not checkable by anyone outside that platform, and the ledger says so rather than assuming it.

The threshold is explicit and small: publish the 604 vectors in any exact or decimal form and the row is decided in minutes.

5 Scope

No bound is improved here and no configuration was searched for. Upper bounds, which come from semidefinite programming, are untouched; this paper concerns lower-bound witnesses only. Where a producer had already verified its own configuration exactly, the contribution is independence: different code, a different language, different arithmetic, and no shared line with the producer.

Reproduction

Every constant in this paper is interpolated from a certificate record in the `cert-machine` repository, so the document cannot quote a number the machine did not certify. The instrument is `instruments/kissing`; the ledger is `certs/kissing-ledger.json`; the pinned coordinates are in `corpus/kissing` with the upstream digest of each source recorded. Running `node instruments/kissing/battery.js` re-derives the D_4 and E_8 calibrations from their definitions, re-certifies one 604-point configuration live, and requires every red control to fire, among them the mixed-sign $\sqrt{2}$ comparator and the exactness of decimal-literal parsing.

References

- [1] A. M. Odlyzko and N. J. A. Sloane. New bounds on the number of unit spheres that can touch a unit sphere in n dimensions. *J. Combin. Theory Ser. A* 26 (1979), 210–214.
- [2] O. R. Musin. The kissing number in four dimensions. *Ann. of Math.* 168 (2008), 1–32.
- [3] M. R. Best. Binary codes with a minimum distance of four. *IEEE Trans. Inform. Theory* 26 (1980), 738–742.